Weight Quantization and the Stability of Turkish Grammatical Judgments in EuroLLM

Abstract

Language models used in Turkish text research should be evaluated for linguistic stability as well as computational cost. This study examines how weight quantization changes EuroLLM-1.7B's grammatical preferences on 1,000 deduplicated TurBLiMP minimal pairs across four categories. The same sentences are scored with a local FP16 reference and MLX affine 8-bit and 4-bit configurations. Mean token log probability defines the primary decision rule. FP16, Q8, and Q4 achieve accuracies of 91.2%, 91.5%, and 88.4%, respectively. The Q8 difference of +0.3 percentage points is not statistically significant and does not establish equivalence. Q4 loses 2.8 percentage points (Holm-adjusted $p = .000469$), with 42 correct-to-incorrect and 14 incorrect-to-correct transitions. Subject agreement accounts for 18 of the 28 net lost decisions. Summed scoring increases the Q4 deficit to 3.9 percentage points. Exploratory analyses do not show a monotonic increase in reversals with subword fragmentation. Measured peak MLX allocator memory is 63.8% lower for Q4. For computational social research, the contribution is a reproducible diagnostic of a linguistic component: aggregate accuracy, decision stability, category-specific errors, and scoring choices must be assessed together. The findings establish an accuracy–memory trade-off within the tested setting; their implications for substantive text coding require application-specific evidence.

Keywords: weight quantization, Turkish, measurement validity, computational social science, morphosyntax, reproducibility.

1. Introduction

Computational social research often turns textual material into observations. The credibility of the resulting analysis depends partly on the language-processing instruments used to produce those observations. Grimmer and Stewart (2013) emphasize that automated text analysis requires extensive, problem-specific validation. A practical question arises when a researcher replaces a model with a version that uses less memory: which linguistic decisions remain stable, and which change? This study addresses that question for Turkish through a controlled comparison of weight precision.

The methodological problem is instrument substitution. Lower computational requirements may make a model easier to run locally, but do not establish that it produces interchangeable observations. Agreement with the original model and agreement with an external reference are separate properties. We examine both through a minimal-pair task, in which the model ranks a benchmark-preferred sentence against a minimally different alternative. The measured outcome is grammatical preference

on those contrasts. It is a linguistic diagnostic relevant to the preparation of text-analysis methods, rather than a direct measurement of a social concept.

Running language models on local hardware requires sufficient memory for both the weights and the computations performed during inference. Representing weights at lower precision reduces their storage requirements. Whether the resulting model preserves the grammatical distinctions made by its reference counterpart is a separate empirical question. Multilingual evaluations have shown that the effects of quantization vary across languages and tasks and that automatic metrics can underestimate the degradation observed by human evaluators (Marchisio et al., 2024).

In Turkish, grammatical relations such as agreement, case selection, and subordination are often expressed through morphology. A single word can contain several morphemes, while a language model's tokenizer can place boundaries that do not coincide with morpheme boundaries. Sentence length and the distribution of morphemes can therefore affect the interpretation of minimal-pair scores (Başar & Bisazza, 2026). A decline in accuracy alone does not establish which grammatical distinction has become harder to recognize.

We evaluate three precision configurations of EuroLLM-1.7B (Martins et al., 2024) on the same Turkish minimal pairs from TurBLiMP (Başar et al., 2025). The overarching methodological question is how the choice of weight precision changes the linguistic observations obtained from a fixed Turkish benchmark. RQ1 asks how 8-bit and 4-bit quantization change accuracy and individual decisions relative to FP16. RQ2 asks whether the conclusions remain stable across grammatical categories, sentence-scoring conventions, and sensitivity analyses for related or disputed pairs. RQ3 asks what accuracy–memory trade-off is observed under the measured local workload. An exploratory analysis examines associations between subword fragmentation and Q4 reversals.

The contribution is a documented procedure for evaluating a linguistic component before substituting one model configuration for another. It combines paired decisions, benchmark agreement, grammatical categories, scoring sensitivity, and workload-specific resource measurements. Its relevance to computational social science lies in making the construction and stability of textual observations inspectable. Application validity remains a separate question: the present benchmark cannot establish the quality of sentiment, stance, or other substantive codes.

2. Related Work

2.1. Low-precision Inference

Quantization schemes with the same nominal bit width can implement different numerical operations. LLM.int8() retains a higher-precision computation path for outlier features while using 8-bit matrix multiplication for most values (Dettmers et al., 2022). GPTQ uses approximate second-order information for post-training weight quantization (Frantar et al., 2023). AWQ uses activation statistics

to identify scaling factors that protect salient weights (Lin et al., 2024). SmoothQuant redistributes quantization difficulty between activations and weights to support 8-bit computation for both (Xiao et al., 2023). The present comparison holds the MLX affine scheme and group size fixed and varies the weight bit width. No further training or task-specific fine-tuning was performed.

2.2. Multilingual and Turkish Evaluations

Marchisio et al. found that quantization affected languages differently across multilingual models (Marchisio et al., 2024). Their results motivate language-specific evaluation but do not establish a ranking of Turkish relative to other languages. Bayram et al. examined quantization in Turkish classification and summarization tasks alongside system efficiency (Bayram et al., 2025). Our outcome differs from those task scores: we track whether a model changes its preference between two sentences and whether each change agrees with the benchmark label. Differences in tasks and scoring preclude a direct ranking across the studies.

2.3. Minimal Pairs and Morphology

BLiMP uses minimally contrasting sentences to probe language models' sensitivity to grammatical distinctions (Warstadt et al., 2020). TurBLiMP extends this approach to 16 Turkish linguistic phenomena, including constructions involving flexible word order and morphological subordination (Başar et al., 2025). Başar and Bisazza investigated how morpheme counts, subword counts, length, and tokenizer alignment with morphological boundaries affect such evaluations (Başar & Bisazza, 2026). We treat minimal-pair accuracy as a measure of sensitivity to the selected contrasts. Our token-to-word ratio is a sentence-level proxy for fragmentation and does not measure morpheme counts.

2.4. Validation in computational text research

Adcock and Collier (2001) distinguish the conceptual and observational levels involved in measurement and emphasize the contextual specificity of validity claims. This provides a useful starting point for assessing language-model outputs: the meaning of a target concept must be specified before a score can be interpreted as its indicator. For this study, the operational target is preference for the benchmark-designated grammatical alternative, not general communicative competence or comprehension of a social claim.

Gilardi et al. (2023) assess language-model annotation against human judgments in application-specific text tasks. Their approach illustrates why evidence should correspond to the outcome being claimed. The present evaluation instead uses a small base model, sentence likelihoods, and grammatical contrasts. The two settings address different levels of validation. A grammatical diagnostic can reveal sensitivity to a computational change, while an application-level evaluation determines whether the outputs support the intended substantive analysis. Keeping these levels separate allows linguistic evidence to inform research design without overstating its reach.

3. Measurement Framework

3.1. Weight Approximation and Preference Margins

Groupwise affine quantization represents weights approximately using low-bit integer codes and a scale and offset shared within each group (MLX Contributors, n.d.). Approximation error can alter the probabilities assigned to sentence tokens without changing the preferred sentence. Conversely, a small shift can reverse a preference when the original scores are close. For each pair, the margin is the score of the benchmark-preferred sentence minus the score of its alternative. A positive margin indicates agreement with the benchmark. A change of sign indicates a reversal.

An association between reversals and small absolute baseline margins is partly a consequence of this decision rule. A point close to zero requires a smaller displacement to cross the boundary. Such an association can identify vulnerable decisions without identifying the representation of a particular suffix as the cause.

3.2. Tokenization and Morphosyntax

With a fixed tokenizer, changing weight precision leaves the token sequence of a given sentence unchanged. Matching token identities across conditions is therefore a validity check. The empirical question is whether an existing characteristic of those sequences helps explain which decisions change. A suffix may be represented by different subword units in different contexts. The available records permit a test using a fragmentation proxy but do not identify morpheme boundaries or changes in internal representations.

3.3. Levels of evidence and permissible inference

Three relationships define the evaluation. Configuration stability concerns whether changing weight precision changes the preferred sentence for a fixed pair. Benchmark agreement concerns whether that preference matches the supplied grammatical label. Application validity concerns whether a model output measures the construct of interest in a research corpus (Adcock & Collier, 2001). The first two are observed in this experiment. The third sets the boundary of interpretation: grammaticality scores are not operational definitions of substantive social variables.

This distinction matters because identical errors can produce perfect stability. Conversely, a reversal can improve agreement with the benchmark. We therefore report both directions of change and retain the underlying continuous margins. A model that reproduces the reference is not necessarily a valid instrument, and a model that differs from it is not necessarily worse on every item. These statements follow from the definitions of stability and accuracy; they do not assume that FP16 provides ground-truth language judgments.

The audit also distinguishes the measurement rule from the computational configuration. The rule specifies how token probabilities become a sentence score and how two scores become a binary decision. The configuration specifies the model weights, tokenizer, numerical representation, and runtime. Holding one fixed while varying the other makes their contributions observable. When both scoring conventions and weight precision change, their effects should not be collapsed into an unspecified change in model quality.

These distinctions make the methodological question testable with the existing paired design. A replacement can preserve aggregate accuracy while changing individual observations, or reduce accuracy mainly within a particular grammatical category. Either pattern would be obscured by reporting a single overall score. The comparison therefore evaluates the distribution of changes as well as their net direction. This approach is useful when linguistic decisions form part of a wider research pipeline, even though the benchmark itself does not measure the final substantive construct.

4. Data and Task

We used four TurBLiMP categories: subject agreement, transitive argument structure, nominalization, and irregular forms. Each source file contained 1,000 pairs. We screened for empty sentences, identical alternatives, and exact duplicate pairs. No invalid rows or cross-category exact duplicates were found. Removing 338 within-category duplicates left a pool of 3,662 distinct pairs. Using Python's random.Random(42) and a fixed category order, we sampled 250 pairs from each category. See Table 1.

Table 1

Sampling Frame and Selected Pairs.

Source: Author's calculations from the experimental records.

Category	Raw	Duplicates	Unique	Selected
Subject agreement	1000	66	934	250
Argument structure	1000	165	835	250
Nominalization	1000	55	945	250
Irregular forms	1000	52	948	250
Total	4000	338	3662	1000

A duplicate required an exact match of both sentences within a category.

The source revision recorded in the project manifest was 297de13fb7a0ce524fe32e8b175c6b5255d66960. The SHA-256 digest of the selected JSONL sample was bcb2c7f25475f8290d1511d742e1e9f8f515fd29df11dd3e3f64f2f53bcb4835. Resampling the local

source files reproduced the content and order of the selected pairs. The digest documents file integrity. Linguistic label validity requires separate evidence.

The sample contained 1,000 unique pair identifiers and 1,000 distinct sentences on each side of the contrast. Equal category sizes make micro-accuracy identical to the unweighted mean of the four category accuracies. These weights describe the benchmark design rather than the prevalence of the phenomena in naturally occurring Turkish. Ellipsis, morphological tagging, and inflected-word generation were outside the evaluation scope.

Each sentence was scored separately. A decision was counted as correct when the benchmark-preferred sentence received the higher score. The task required neither free generation nor a verbal grammaticality judgment. The preferred and dispreferred sentences averaged 13.415 and 13.464 tokens, respectively, with ranges of 6–28 and 6–29. No truncation was required within the 4,096-token context window. Each configuration scored 26,879 sentence tokens, excluding the beginning-of-sequence context.

5. Methods

5.1. Model Configurations

EuroLLM-1.7B is a multilingual decoder-only base model that includes Turkish (Martins et al., 2024; UTTER Project, n.d.). The local configurations shared the same architecture and tokenizer. The model has 24 layers, a hidden dimension of 2,048, 16 attention heads, and eight key/value heads. Q8 and Q4 used affine weight quantization with a group size of 64. No chat template, conversation history, demonstrations, or task-specific adaptation was used.

The reference is designated FP16 on the basis of the F16 tensors in the local weight file. The BF16 field in the source configuration describes a different aspect of the model. The quantized files contain packed U32 weights and F16 auxiliary tensors. The Q4 designation therefore refers to weight representation and does not imply that all arithmetic was performed at four bits. Project documentation records the Hugging Face download reference as refs/pr/5 and describes separate Q8 and Q4 conversions from FP16. That reference is mutable; the exact resolved upstream commit and complete conversion commands were not archived, which limits reconstruction of the weight-preparation stage.

5.2. Sentence Scoring

Sentences were tokenized without automatically adding special tokens. One BOS token supplied the initial context. No EOS token was appended or scored. Each observed sentence token was scored conditional on the preceding context. Logits were cast to float32 before log-softmax, and the log probabilities of the observed tokens were averaged. This cast applied to score calculation rather than the entire model.

$$S_q(s) = \frac{1}{T}\sum_{t=1}^{T} \ln p_q(x_t \mid BOS, x_{<t}) \quad (1)$$

Here T is sentence length in tokens and q denotes the configuration. For pair i, the preferred and dispreferred sentences are denoted by s^+ and s^-. Accuracy is the proportion of strictly positive margins.

$$m_{q,i} = S_q(s^+) - S_q(s^-), A_q = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}(m_{q,i} > 0) \quad (2)$$

No exact ties occurred. Summed log probability was retained as a sensitivity analysis. Scores cover complete sentences, so a reversal cannot be attributed specifically to the probability of the critical suffix.

5.3. Paired Inference

The statistical unit was the minimal pair. We compared each quantized configuration with FP16 using a two-sided exact McNemar test (McNemar, 1947). Let d count correct-to-incorrect transitions and u count incorrect-to-correct transitions. Under the null hypothesis, conditional on d + u discordant pairs, the number of transitions in one direction follows a binomial distribution with probability 0.5. We applied Holm adjustment (Holm, 1979) to the two primary comparisons at a significance level of .05. No time-stamped preregistration was documented.

Individual accuracy intervals used the 95% Clopper–Pearson procedure (Clopper & Pearson, 1934). For the paired difference, we calculated separate 97.5% Clopper–Pearson intervals for u/N and d/N. Subtracting the upper bound for degradation from the lower bound for improvement, and vice versa for the opposite endpoint, gives a conservative interval with at least 95% coverage under the independent-pair sampling model. These intervals apply to individual contrasts. Holm adjustment was applied to p values.

5.4. Dependence Among Related Pairs

Distinct pairs may share a linguistic core. Within each category, we grouped pairs whose normalized alternatives matched after removing shared leading words from a specified list: ama, ancak, ardından, artık, aslında, ayrıca, belki, bugün, burada, evet, fakat, hadi, hemen, işte, sadece, sonra, ve, yalnız, yani, yine, çünkü, öncelikle, özellikle, and şimdi. The resulting 776 clusters approximate related examples. Their membership has not been validated against the original generation templates.

We used category-stratified bootstrap resampling (Efron, 1979) of complete pairs and, separately, of complete approximate clusters. Each procedure used 20,000 replicates for the overall accuracy contrast, with seeds 20261028 and 20261029, respectively. Categories retained equal weights. Dependence beyond that captured by the clustering rule may remain.

5.5. Exploratory Fragmentation Analysis

For each pair, the fragmentation proxy was the arithmetic mean of the two token-to-word ratios.

$$F_i = \frac{1}{2}\left(\frac{T^+}{W^+} + \frac{T^-}{W^-}\right) (3)$$

T and W denote token and word counts. Words were counted as Unicode letter or digit sequences, preserving internal apostrophes and treating hyphens as boundaries. Standalone punctuation contributed to the token count but not the word count. The outcome of the logistic regression was any reversal between FP16 and Q4, regardless of direction.

The unadjusted model contained fragmentation alone. The main adjusted model also included absolute FP16 margin, mean word count, and category, with argument structure as the reference. All predictors, including category indicators, were standardized. We added a ridge penalty equal to λ/2 times the sum of squared slopes to the total negative log likelihood and left the intercept unpenalized. The main penalty was λ = 1, with 0.1 and 10 examined for sensitivity. No predictive tuning or held-out classifier evaluation was performed.

Pair and cluster resampling each used 2,000 replicates, with seeds 20261029 and 20261030. We refitted the scales and coefficients in each replicate and normalized category weights to their original sizes. Odds ratios were expressed per original-sample standard deviation of fragmentation. Percentile intervals describe uncertainty in the penalized estimates and do not correct regularization bias or multiplicity across exploratory models. Q8 had only three reversals and was not modeled separately.

5.6. Audit procedure and interpretation rules

The audit treats the pair identifier as the join key across precision settings and verifies that the same token sequences were scored. It retains continuous scores, binary decisions, and benchmark labels so that a reader can separate reference agreement from label agreement. Overall accuracy is accompanied by transition counts, category-specific results, and normalization sensitivity. The resource measurements are attached to the exact scoring workload rather than generalized to all deployments.

Interpretation follows the estimand and measurement scope. The paired accuracy contrast describes agreement with the benchmark, whereas the reversal count describes stability relative to FP16. A nonsignificant contrast does not establish equivalence, and category-specific losses do not identify demographic effects. Resource findings refer to the recorded workload and the memory measurement used. Together, these rules prevent accuracy, stability, and computational cost from becoming interchangeable indicators of overall model suitability.

6. Experiments

The experiment ran on an Apple M2 MacBook with 16 GiB of unified memory, macOS 26.4.1, Python 3.12.15, MLX 0.32.4, and MLX-LM 0.32.2. Batch size was one sentence. We processed 20 chunks of 50 pairs and recorded scoring times after loading and warm-up. The additional analyses used NumPy 2.5.4, SciPy 1.18.1, and Matplotlib 3.11.2.

The saved experiment contains 3,000 pair–configuration records and 6,000 sentence scores. Consistency checks covered identifiers, categories, token counts and digests, and score arithmetic. The maximum absolute discrepancy between the stored mean and the sum divided by token count was 4.24×10^{-7}. Reanalysis of the saved scores reproduced the reported main accuracy and paired-test results. These checks assess record integrity and computational consistency; they do not add new test items.

Mean token log probability defined the primary analysis. Summed scores, category breakdowns, pilot exclusion, label sensitivity, margin distributions, and fragmentation models provided secondary or exploratory analyses. The pilot included five pairs per category drawn from the main sample. A complete rerun in separate processes reproduced every decision. It did not add independent observations. Background load was uncontrolled during the rerun, limiting performance comparisons.

7. Results

7.1. Overall Accuracy and Transitions

FP16, Q8, and Q4 produced 912, 915, and 884 correct decisions, respectively. Q8 retained every correct FP16 decision and corrected three errors. The small positive difference did not establish superiority. Equivalence or noninferiority would require a prespecified acceptable loss. See Table 2.

Table 2

Primary Accuracy and Paired Contrasts against FP16.

Source: Author’s calculations from the experimental records.

Model	Correct / N	Accuracy % [95% CI]	Δ pp	Paired 95% CI, pp	Holm p
FP16	912/1000	91.2 [89.27, 92.88]	—	—	—
Q8	915/1000	91.5 [89.60, 93.15]	+0.3	[−0.390, 0.971]	.250
Q4	884/1000	88.4 [86.25, 90.32]	−2.8	[−5.151, −0.416]	.000469

pp = percentage points. Accuracy intervals are exact binomial intervals. Difference intervals use the conservative paired procedure.

Q4 reversed 56 decisions (5.6%). Forty-two were degradations and 14 were improvements, yielding a net loss of 28 correct decisions. FP16 and Q4 were both correct for 870 pairs and both incorrect for 74.

The error count rose from 88 to 116, a relative increase of 31.8%. All three configurations agreed on 870 correct and 72 incorrect decisions.

For Q4, the category-stratified pair bootstrap interval was [−4.30, −1.40] pp and the approximate-cluster interval was [−4.45, −1.21] pp. For Q8, bootstrap lower bounds of zero reflect the absence of observed degradations: resampling cannot generate a transition type absent from the observed sample. The conservative interval is therefore the primary basis for interpretation.

7.2. Category Differences

Subject agreement accounted for 18 of the 28 net lost decisions (64.3%). Argument structure and nominalization contributed four and seven net losses, respectively, while irregular forms contributed one net gain. See Table 3.

Table 3

Category Accuracy and Q4 Transitions (250 Pairs per Category).

Source: Author's calculations from the experimental records.

Category	FP16 %	Q8 %	Q4 %	Q4 Δ pp	d / u
Subject agreement	88.4	88.8	81.2	−7.2	23 / 5
Argument structure	93.2	93.2	91.6	−1.6	6 / 2
Nominalization	84.0	84.8	81.2	−2.8	13 / 6
Irregular forms	99.2	99.2	99.6	+0.4	0 / 1

d and u denote degradations and improvements relative to FP16.

Across a separate exploratory Holm family of eight category comparisons, the adjusted p value for Q4 subject agreement was .00730. The other comparisons did not reach the .05 threshold. Differences in significance across categories do not establish an interaction. Irregular forms were already near ceiling at 99.2% under FP16, leaving only two errors available for improvement.

7.3. Baseline Margins

Mean absolute margin shifts relative to FP16 were 0.00838 for Q8 and 0.10509 for Q4. Pearson correlations with FP16 margins were .99939 and .90822, respectively. Close agreement across the full distribution can coexist with sign changes near zero. Across successive absolute FP16 margin quartiles, Q4 reversed 46, 8, 1, and 1 decisions. The lowest quartile accounted for 82.1% of reversals. Its reversal rate was 18.4%, compared with 1.33% among the remaining 750 pairs. See Figure 1.

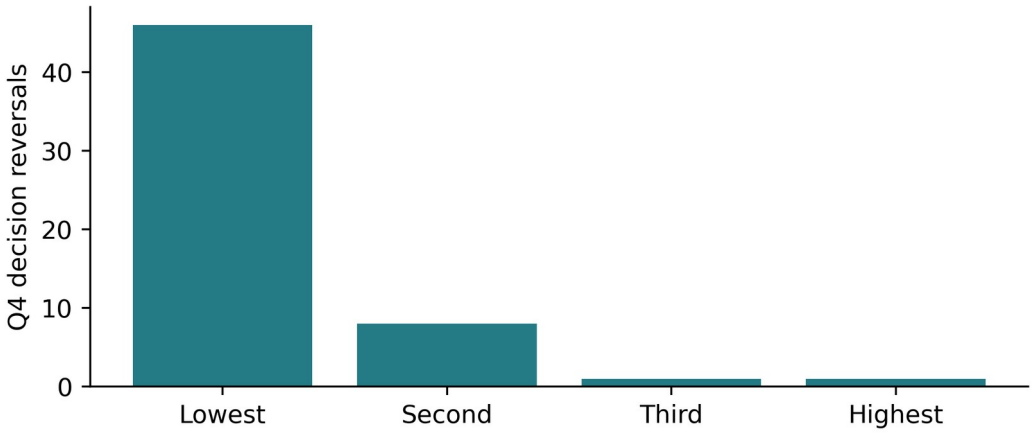


Figure 1

Q4 Reversals by Absolute FP16 Margin Quartile (250 Pairs Each).

Source: Author's calculations from the experimental records.

7.4. Scoring and Sample Sensitivity

Under summed log probability, accuracy was 94.6% for FP16, 94.8% for Q8, and 90.7% for Q4. The Q4 contrast increased in magnitude to −3.9 pp. Switching between mean and summed scoring changed 60 FP16, 59 Q8, and 81 Q4 decisions. In the 626 pairs with equally long token sequences, both scoring rules yielded the same preference for every configuration. See Table 4.

Table 4

Sensitivity of the Q4–FP16 Contrast.

Source: Author's calculations from the experimental records.

Analysis	N	Δ pp	Paired 95% CI, pp	Holm p
Primary mean score	1000	−2.800	[−5.151, −0.416]	.000469
Summed score	1000	−3.900	[−6.176, −1.576]	3.04×10^{-7}
Pilot excluded	980	−2.755	[−5.137, −0.340]	.000711
Eight disputed pairs excluded	992	−2.520	[−4.747, −0.262]	.000940
Predicate screen excluded	983	−2.543	[−4.790, −0.265]	.000940
Subject agreement excluded	750	−1.333	[−3.656, 1.016]	.174317

Holm adjustments apply to the two FP16 comparisons within each analysis.

Excluding the pilot or potentially ambiguous nominalizations preserved the negative Q4 contrast. Excluding subject agreement reduced it to −1.333 pp (adjusted p = .174), illustrating sensitivity to category composition. These exclusions cannot establish label validity in the remaining sample.

7.5. Linguistic Examples

A purposive review covered 20 pairs comprising 12 Q4 degradations, five Q4 improvements, and three Q8 improvements. The review was extended to all 58 unique pairs on which either quantized configuration differed from FP16. The preliminary judgments were AI-assisted. No independent blinded human annotation or inter-annotator agreement was obtained.

For subject_agreement:0532, the preferred sentence was “Ancak o bu projeye çok emek verdi.” The alternative substituted sen (‘you’) for o (‘he/she’) while retaining the third-person predicate verdi. The FP16, Q8, and Q4 margins were 0.301963, 0.304616, and −0.062743. For argument_structure_transitive:0850, “Hadi bu filme bayılıyoruz.” contrasts with “Hadi bu filmi bayılıyoruz.” The verb calls for the dative form filme under the intended reading. The corresponding margins were 0.039436, 0.041425, and −0.167631.

In nominalization:0962, the label favors “Ve şirketin yatırım kararlarını hızla almasına çoğu çalışan şaşırdı.” over the alternative with aldığına. Margins were −0.017708, −0.016909, and 0.236379. Event-oriented and fact-oriented interpretations may license both complements, so Q4’s agreement with the label is an ambiguous linguistic improvement. Of the 58 reviewed pairs, eight raised similar concerns, three were context-sensitive, and five required expert review. The selection does not support a corpus-wide estimate of labeling errors.

7.6. Subword Fragmentation

The sample quartile thresholds were 1.857143, 2.083333, and 2.339286 tokens per word. Ties at a threshold remained in the lower group. The group sizes were 256, 247, 247, and 250. Reversal rates were 5.86%, 4.45%, 8.91%, and 3.20%, with no monotonic increase. The highest-minus-lowest contrast was −2.66 pp (approximate-cluster bootstrap 95% interval [−6.50, 1.41]). See Figure 2.

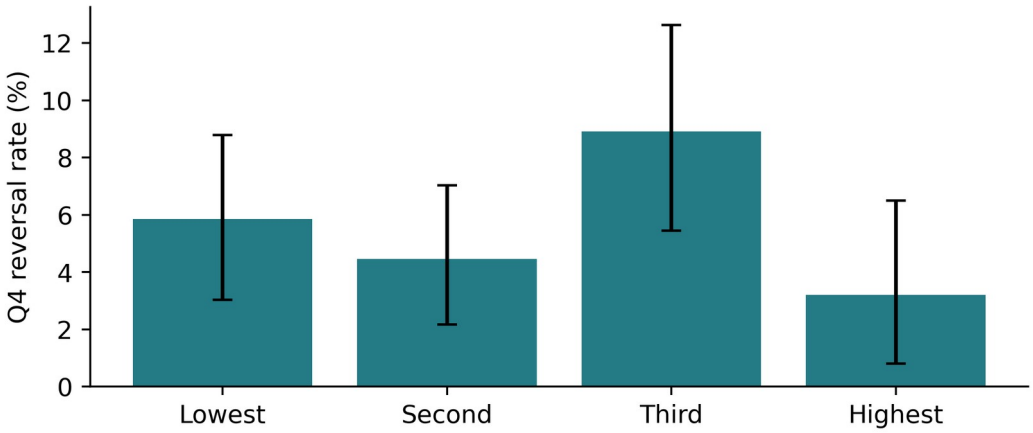


Figure 2

Q4 Reversal Rates by Fragmentation Quartile. Bars Represent 15/256, 11/247, 22/247, and 8/250 Pairs. Error Bars Show Category-stratified Approximate-cluster Bootstrap 95% Intervals.

Source: Author's calculations from the experimental records.

Table 5

Fragmentation and the Odds of a Q4 Reversal.

Source: Author's calculations from the experimental records.

Model	OR per SD	Pair bootstrap 95% CI	Cluster bootstrap 95% CI
Unadjusted	.789	[.611, .982]	[.592, 1.027]
Margin, category, word length	.735	[.548, .932]	[.540, .946]
Margin, category, token length	.900	[.664, 1.193]	[.649, 1.230]

One SD corresponds to 0.3977 tokens per word. Intervals are exploratory and unadjusted for multiple comparisons.

Adjustment for within-pair fragmentation differences yielded an OR of .742 (cluster interval [.544, .952]). Using log absolute margin gave .737 [.546, .953]. Penalties of 0.1 and 10 produced point estimates of .716 and .817, without additional intervals. The negative adjusted association weakened when token count replaced word count. These results offer no evidence for a general protective effect of fragmentation. An odds ratio describes a change in odds, and the outcome includes both improvements and degradations. See Table 5.

7.7. Resource Measurements

In the original run, scoring took 20.4% less time for Q8 and 22.6% less for Q4 relative to FP16. Measured peak MLX allocator memory fell by 63.8% for Q4, and weight-file size fell by 71.9%. See Table 6.

Table 6

Resource Measurements in the Original Run.

Source: Author's calculations from the experimental records.

Measure	FP16	Q8	Q4
Scoring time (s)	123.04	97.88	95.20
Peak MLX allocator memory (GB)	3.416	2.068	1.237
Peak process RSS (GB)	1.247	1.839	1.573
Weight files (GB)	3.314	1.761	0.932

GB denotes 10^9 bytes. Allocator memory, resident set size, and storage have distinct measurement scopes.

Timing includes tokenization, model computation, score extraction, record preparation, and cache clearing between pairs. Loading, warm-up, file writing, and pauses between chunks were excluded. The Q8–Q4 difference of 2.68 seconds represents one run. In the verification run, FP16, Q8, and Q4 took 127.08, 104.86, and 98.85 seconds, respectively, with identical decisions. The 20 chunks contain different sentences and cannot serve as 20 independent performance replicates. Energy consumption, temperature, memory pressure, and swap were not measured concurrently.

8. Discussion

8.1. Stability of the linguistic instrument

The Q4 configuration reduced measured memory use at a cost in accuracy. Q8 remained close to the reference on this sample. The acceptable loss for deployment depends on the intended application, and no practical tolerance was specified before analysis. The observed trade-off consequently applies to the reported workload.

Subject agreement contributed disproportionately to the overall decline. This pattern cannot be attributed specifically to damaged person-suffix representations: lexical choices, template structure, and baseline difficulty vary across categories. No token- or layer-level intervention isolated an internal mechanism. Baseline margins help describe vulnerable examples, although they are not calibrated confidence probabilities or a validated routing criterion.

The fragmentation results complement morphology-aware evaluations (Başar & Bisazza, 2026). A word split into several tokens need not represent a difficult grammatical contrast. Punctuation contributes to the numerator of the proxy, and length enters both the ratio and the adjustment model. The token-length sensitivity interval includes the neutral value, which limits any interpretation of the negative slope as a robust linguistic effect.

Mean and summed scores yielded different absolute accuracy levels and different sizes of the Q4 deficit. Reporting both while retaining the original primary rule makes the effect of normalization visible. Comparisons with other minimal-pair studies should account for their scoring conventions.

8.2. Implications for computational social research

Instrument substitution should be assessed at the level of the observations it changes. The observed Q4 loss is not distributed uniformly: some decisions improve, others deteriorate, and subject agreement contributes most of the net decline. An aggregate score alone would conceal this pattern. For researchers building Turkish text-analysis pipelines, category-specific diagnostics can identify which linguistic distinctions need scrutiny when precision changes. The results do not demonstrate a corresponding error rate in an untested application, but they show why nominally similar model configurations should be evaluated rather than assumed interchangeable.

The normalization analysis also locates part of the measurement uncertainty outside the weight representation. On the same sentences, mean and summed log probability yield different accuracy levels and different estimates of the Q4 deficit. Neither normalization can be selected solely because it gives a favorable comparison. A research workflow should specify its scoring rule before the main evaluation, retain an interpretable alternative as a sensitivity check, and explain what construct the resulting score is intended to measure. Here, both rules identify a negative Q4 contrast, while disagreeing about its size.

Benchmark labels introduce another source of uncertainty. The nominalization examples show that a change toward the preferred label may depend on the reading assigned to a sentence. The AI-assisted review is diagnostic and purposively selected, so it cannot establish a corpus-wide labeling-error rate. Accordingly, configuration stability can be measured directly from paired decisions, while claims about linguistic correctness remain conditional on label validity. This distinction is particularly relevant when a convenient benchmark is used to justify the selection of an instrument for a different corpus.

8.3. Resource requirements and local research workflows

Memory is an operational constraint in local research workflows. The Q4 configuration used less measured allocator memory, while Q8 combined a smaller memory footprint with an accuracy close to FP16 in this sample. These results provide workload-specific information for choosing which

configurations to examine further. They do not measure institutional access, adoption, monetary expenditure, or energy consumption. The relevant methodological trade-off is therefore between a recorded resource reduction and a recorded change in grammatical decisions, rather than an inferred social benefit.

The Q8 result offers a useful comparison but not a deployment recommendation. Its accuracy is close to FP16 in this sample and its weight file and allocator memory are smaller. An equivalence or noninferiority claim would require an application-specific acceptable loss and a suitable design. Choosing such a tolerance after observing the present result would change the inferential question. Researchers can use the reported intervals to plan a future evaluation, but cannot retrospectively convert the superiority test into a prespecified equivalence test.

8.4. Reporting a precision comparison in text research

A usable precision comparison should identify its unit of observation, reference labels, numerical configuration, and decision rule. Reporting the same items across configurations makes individual changes traceable. Retaining continuous margins explains how a stable distribution can contain vulnerable decisions near the boundary. Disaggregated results expose category concentration, and sensitivity analyses indicate how scoring or sample composition affects the conclusion. The replication materials retain these components so that readers can inspect the path from sentence scores to reported findings.

For computational social research, this procedure contributes to transparent instrument evaluation. The 2.8-percentage-point Q4 decline is an estimate for this benchmark and scoring rule; it is not a transferable correction factor for other text variables. Researchers can use the diagnostic to document a model-selection decision and its limits. Whether a selected configuration is suitable for a substantive research question must be judged against that question's own evidence requirements, as emphasized in the validation literature (Grimmer & Stewart, 2013).

9. Limitations

We evaluated one base model, one quantization scheme, one language, and four categories. Other model scales, monolingual Turkish models, and calibration-based methods may behave differently. No cross-language control permits attribution to Turkish or to agglutinative languages generally. Overlap between test sentences and pretraining data was not investigated.

Balanced category sampling does not represent natural usage or the entire benchmark. Exact deduplication leaves related templates, and the clustering heuristic captures only selected initial-word variants. Exploratory AI-assisted review does not substitute for blinded judgments by independent Turkish-language experts. Exclusion analyses address only the label issues identified.

Whole-sentence rankings do not measure inflection generation, lemmatization, or morpheme tagging. The fragmentation proxy lacks gold morpheme boundaries. The logistic model estimates a common linear log-odds slope without category interactions or nonlinear terms. Sparse events and regularization limit coefficient interpretation.

The pilot overlaps the main sample, the study was not preregistered, and multiplicity was not controlled across all exploratory analyses. The replication package supports reanalysis of the saved scores, but incomplete model-conversion provenance limits reconstruction from the upstream weights. Decision reproducibility on the same inputs is distinct from replication on new items. Timing comparisons were not based on repeated full runs with randomized configuration order.

The empirical scope is a Turkish grammatical benchmark. The study does not test a substantive social-science variable or estimate how quantization changes a downstream research conclusion. Its methodological contribution is confined to the evaluation of a linguistic component used in preparing text-analysis instruments. This boundary should be retained when using the findings to motivate broader applications.

10. Conclusion

Across 1,000 matched Turkish minimal pairs, Q8 differed from FP16 by +0.3 pp without establishing superiority. Q4 differed by −2.8 pp, a statistically significant decline after Holm adjustment. Its 56 reversals included changes in both directions, concentrated in low-margin items. Subject agreement accounted for most of the net decline.

The exploratory analysis did not support an increase in decision changes with greater fragmentation. The inverse conditional association depended on the definition of length. Alongside a 63.8% reduction in measured peak MLX allocator memory, these findings quantify an accuracy–memory trade-off for the tested model, sample, and scoring procedure. The scope of this finding is bounded by benchmark-label uncertainty, related templates, incomplete conversion provenance, and the limited performance-measurement design.

For computational research using Turkish texts, model configuration and measurement procedure should be reported together. Overall accuracy, item-level stability, grammatical categories, and resource scope each contribute different evidence about a potential substitution. The present comparison makes that evidence inspectable through a fixed sample and saved scores. It supports transparent evaluation of a Turkish linguistic component while reserving claims about substantive research validity for the relevant application.

11. Research Transparency

11.1. Data and Code Availability

The original TurBLiMP benchmark is available at https://github.com/ezgibasar/TurBLiMP (Başar et al., 2025). This experiment uses source revision 297de13fb7a0ce524fe32e8b175c6b5255d66960 and the sample digest reported in Section 4. The supplementary replication package provides the selected sample, source CSVs, saved scores, analysis code, model configuration files, figure assets, and a SHA-256 inventory. It supports reanalysis without model inference and excludes model weights. The author's public project repository contains the core experiment code and saved results; its identifying URL is provided separately to the editor to preserve anonymous review. The supplementary package additionally supplies the selected data and exploratory analyses. The recorded download reference and incomplete conversion history constrain reconstruction from the original weights.

11.2. AI Use Declaration

OpenAI Codex assisted with Python code development, analysis checks, literature searching, English-language revision, and document preparation in this project. The manuscript revision of 7 November 2026 used Codex with GPT-6.1; exact model identifiers for earlier assistance were not retained. Preliminary linguistic comments on selected sentence pairs were AI-assisted and are treated as exploratory rather than independent human judgments. Experimental predictions are the likelihood scores produced by the EuroLLM configurations described in Sections 5 and 6; Codex outputs are not additional benchmark observations. The sole human author is responsible for the research, submitted text, references, and interpretation.

11.3. Ethics and consent

This study analyzes existing benchmark sentences and model likelihood scores. It did not recruit human participants, collect participant responses, or analyze identifiable personal records. Participant consent is therefore not applicable to this secondary benchmark analysis. No ethics-committee approval or formal exemption was obtained. The study reports no human-annotation experiment.

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